

The Headline Shift: Measuring Ideology and Sentiment in U.S. Online News Headlines Over Time

Sophie Berrocal, Sam DeLong, Annie Dong, Emily Koepp

I. Headlines in the Modern Political Media Environment

In a modern media landscape marked by push notifications, short-form media, and inattentive social media scrolling tendencies, the headline has become the entirety of the story for many readers; research shows that news consumers are increasingly failing to click through to read full articles. This could mean that how a headline is worded—and thus the emotions it evokes and the political valence it carries—is the primary information that more and more consumers are gleaning from their traditional print media.

In this context, the sentiment of news headlines carries tremendous implications for understanding modern democratic discourse and the role technology platforms play in shaping the contemporary news landscape in America. Thus, it's increasingly important that political scientists investigate whether American news headlines have truly become more politically polarized and emotionally charged over time. Yet, headlines remain a largely unexplored domain in both political science and natural language processing.

The central machine learning challenge our project addresses is this: can a transformer-based classifier trained on short, noisy headline data reliably detect both ideological lean and emotional intensity simultaneously? And if so, does inference over a decade of headlines reveal time-varying trends in these dimensions that simpler bag-of-words methods fail to capture? These are questions about model capability and generalization, not just about political science — answering them requires careful attention to annotation design, architecture choice, multi-task learning, and evaluation methodology.

II. Literature Review and Our Contributions

A. The Political Economy of Media Bias

Political scientists and communication scholars have long studied media bias, theorizing about agenda-setting, framing effects, and partisan slant. Often, this is understood to be a result of economic goals, where outlets serving ideologically homogeneous audiences encounter commercial incentives to confirm and conform to their readers' ideologies (Gentzkow & Shapiro, 2010; DellaVigna & Kaplan, 2007). However, what has changed in the most recent decade is the degree to which this dynamic is mediated by social platforms and algorithmic recommendation, which amplify ideologically extreme content because it maximizes engagement.

With such new technology transforming the political media environment, the application of NLP and machine learning becomes crucial. Researchers in this field have long been plagued by the challenge of scalability—hand-coding thousands of article headlines for political ideology is both expensive and time-consuming. NLP and machine learning methods thus offer tools for enabling headline research. However, applying these methods to headline text presents distinct

challenges. Since headlines are shorter, more ambiguous, and more strategically constructed than full articles, they're less amenable to methods developed for longer-form text. The existing literature has not fully addressed these challenges, exposing a critical gap in the research.

B. NLP Approaches to Ideology Detection

As we have discussed in STAT3106, early computational approaches to language and ideology detection relied on simple bag-of-words features, handcrafted lexicons, and logistic regression classifiers. These methods have proved relatively effective at separating left- and right-leaning texts, although they are sensitive to domain shift and struggle to capture subtle framing effects.

With the introduction of transformers, beginning with BERT in 2018 and expanding to RoBERTa, DistilBERT, and GPT-family models, there became dramatically improved performance on ideology classification tasks. In a review of automated media bias detection mechanisms, Rodrigo-Ginés et al. (2024) found that transformer-based models now dominate the field. These models can achieve significantly higher accuracy and have the ability to detect more fine-grained bias types. Most importantly, transformer-based models can learn contextual embeddings that capture subtle framing differences. For example, they can distinguish tone differences between “undocumented immigrant” and “illegal alien,” or “pro-life” and “anti-abortion,” while frequency-based methods cannot.

In Spinde et al. (2022), researchers introduced the BABE dataset (Bias Annotations By Experts) and used multi-task transformer fine-tuning to detect media bias in news articles. They found that multi-task learning—jointly predicting bias type, intensity, and presence—outperformed single-task classifiers, a finding that directly informs our decision to jointly model ideology and emotional sensationalism rather than train two completely separate systems.

This result shapes our core architectural decision. Rather than training separate ideology and sensationalism classifiers, we train a single DistilBERT backbone with dual heads, explicitly testing whether joint supervision improves performance on both tasks. Recent work comparing transformer families on bias detection (Bridging Human and Model Perspectives, 2025) further found that RoBERTa achieves the highest alignment with human labels among traditional transformer architectures, informing our choice of backbone for our stronger baseline.

C. The Distinctive Challenge of Headlines versus Full Articles

Most existing bias-detection literature operates on full article bodies, not headlines. This matters for two reasons. First, headlines are far shorter (typically 7–15 words), which means there is less textual signal available to the classifier. Second, headlines are written under different editorial constraints than article bodies — they are optimized for attention and platform shareability in ways that body text is not. D'Alonzo & Tegmark (2022) demonstrated that even simple phrase-frequency approaches could identify which newspaper had published a given

article, with surprisingly high accuracy, but acknowledged that headline-specific models require different feature engineering than full-text classifiers.

The scarcity of headline-specific labeled datasets for ideology classification is a key motivation for our use of active learning. Rather than relying solely on existing article-level ideology labels, we augment existing article-level labels with crowdsourced headline-level annotations from our classmates, generating a novel training signal.

D. Emotional Valence and Clickbait Detection

While the literature on headlines overlaps substantially with clickbait detection research, the two subjects are meaningfully distinct. Whereas clickbait is primarily about deceptive intent to lure readers through misleading cues, headline tone is about emotional manipulation, shaping how readers perceive events through language that triggers strong affective responses regardless of whether the headline is technically accurate. From an ML standpoint, this distinction matters because the two tasks have different label distributions and require different feature sets.

The SENS-HEAD dataset (Goldsmiths University of London, 2024) includes 30,000+ annotated headlines labeled for emotionally laden or sensationalist content. It uses a range of linguistic features including sentiment polarity, syntactic complexity, punctuation patterns, and stop-word ratios. Using ensemble models (XGBoost, CatBoost, Random Forest), researchers achieved a balanced F1 of 0.66—a result that highlights the difficulty of the task and the value of incorporating human annotation loops.

Meanwhile, research using RoBERTa-Large for clickbait detection (Nature Scientific Reports, 2025) found that sentence embeddings with transformer architectures substantially outperform traditional ML baselines, with explainability methods (SHAP/LIME) confirming that the model’s decisions are defined by linguistically meaningful features such as emotional adjectives, verb sentiment, and emotional modifiers rather than spurious correlations.

Thus, the key gap this project addresses is as such: existing work either studies ideology OR emotionality, and typically at the article level, cross-sectionally. Here, we study both simultaneously, treating them as related outputs of a shared representation, testing whether the shared backbone learns more generalizable headline features than single-task training.

E. Temporal Analysis as a Downstream Application

Time-varying analysis of media bias is rare in the computational literature overall. Sinno et al., cited in the media bias systematic review, produced the first time-varying dataset of news articles annotated at the paragraph level, tracking ideological language around specific policy debates. More recently, a study by Lyu et al. (2024) tracked changes headline ideology from 2014 to 2022, but did not control for topic-mix confounders. For example, outlets might have covered more politically polarizing topics in later years, making their headlines appear more extreme without any intentional editorial drift.

Our time-series design explicitly addresses this confound by using the topic tags in the QBIAS dataset to run topic-stratified trend analyses — comparing ideology scores for, say,

economic coverage in 2013 vs. 2022 holding topic constant. This is the core methodological contribution that distinguishes our work from prior efforts.

III. Machine Learning Architecture

A. Task Definition

Component	Definition
Input (x)	A news headline string (typically 7–20 tokens)
Output (y_1)	Ideology label $\in \{\text{Left, Center, Right}\}$ (3-class)
Output (y_2)	Emotional score $\in [0, 1]$ (regression or binary class)
Objective	Cross-entropy (ideology) + Binary cross-entropy (emotion), jointly optimized
Evaluation metrics	Macro F1, Cohen's Kappa (ideology); F1 / AUC (emotion)

B. Data Sources

- Training labels — Ideology
 - We use the AllSides “[OBIAS Media Bias in Search Queries](#)” dataset of 21,000+ headlines labeled left/center/right by political ideology, which also includes topic tags enabling topic-stratified analysis.
- Training labels — Emotional Intensity
 - We generate these via our active learning crowdsourcing interface. Our goal is to have classmates independently rate a shared set of 20–40 headlines on two dimensions—emotional intensity and value-laden language—using a purpose-built comparison interface. Their ratings, combined with agreement thresholds, form the emotional intensity training signal.
- Inference corpus
 - We use a dataset sourced from [Kaggle](#) with 4.5M headlines across the 10 largest U.S. news sites (2007–2022), from which we filter for CNN, Fox News, The Washington Post, and The New York Times in the 2013–2022 window.

C. Active Learning Design

The emotionality annotation task is operationalized as a comparative judgment task rather than an absolute rating task. A scale-based question such as “on a scale from 1–5, is this headline emotional?” is likely to introduce rating-scale bias. To counteract this, we instead present the user with a pair of headlines and ask which of the two headlines “uses more emotional, alarming,

or value-laden language?” This generates a ranking signal that is more reliable and consistent across annotators.

We implement the Bradley-Terry model to convert pairwise preferences into a continuous emotionality score for each headline. The active learning live annotation interface can be found linked in our GitHub repository, or alternatively, via this URL:

<https://unflaggingly-barotropic-oscar.ngrok-free.dev>. Notably, the interface may appear as inactive due to current struggles across multiple permanent interface applications, including HuggingFace, Streamlit, and Supabase.

D. Model Architecture

Our architecture is motivated by a question from the multi-task learning literature: do ideology and emotionality share enough representational structure in headline text that joint training produces better features than training on either task alone? If partisan outlets systematically use more emotionally charged language — which we hypothesize — then a shared encoder trained on both signals should learn richer representations than one trained on a single task.

We thus implement a fine-tuned DistilBERT-base (uncased) backbone with two task-specific heads attached to the [CLS] token representation. As a knowledge-distilled version of BERT, DistilBERT retains approximately 97% of BERT’s language understanding performance at 40% fewer parameters, which is an important practical consideration given that we run inference over 1 million headlines (1,068,680). The ideology head consists of a dropout layer, a linear projection from 768 to 256 dimensions, GELU activation, and a final 3-class softmax output. The emotionality head is intentionally shallower — dropout, linear projection to 128 dimensions, GELU, and sigmoid output — reflecting that emotionality is a simpler binary signal requiring less transformation of the shared representation. The joint training objective is:

$$(1) L = \text{CrossEntropy}(\hat{y}_1, y_1) + \text{BCE}(\hat{y}_2, y_2)$$

Since ideology labels cover all 21,000+ headlines while emotionality labels cover only our annotated subset, we cycle the smaller of the datasets during training and then backpropagate the summed loss through the shared backbone at each step; both tasks regularize each other simultaneously. We compare against three alternatives that together isolate the contribution of each architectural decision:

- **Baseline 1 — TF-IDF + Logistic Regression (ideology only)**
 - This is our strongest non-neural baseline. If DistilBERT fails to substantially outperform it, then the added complexity of the transformer is not justified.
- **Baseline 2 — Fine-tuned RoBERTa-base (single-task and ideology only)**
 - RoBERTa has shown stronger performance on bias classification tasks than BERT-family models (Bridging Human and Model Perspectives, 2025). Running it single-task isolates backbone quality from training paradigm, so any remaining performance difference between this and our primary model is attributable specifically to multi-task learning.

- **Ablation — DistilBERT ideology head only**
 - Identical to our primary model except the emotionality head and its loss term are removed. Any difference in ideology F1 between this ablation and the full model is attributable solely to the joint training signal from emotionality annotations — making this the most direct test of our core architectural hypothesis.

E. Evaluation

Our primary metric for *ideology classification* is macro-averaged F1, which averages F1 equally across all three classes regardless of their frequency. We use this because it is essential given our class imbalance. A classifier that always predicts "left" would achieve 47% accuracy but near-zero macro F1, immediately exposing it. We supplement this with **Cohen's Kappa**, which measures label agreement corrected for chance, providing an interpretable signal of whether the model is genuinely learning or exploiting class priors.

For emotionality classification, we use **binary F1** to measure performance on the sensationalist/not-sensationalist labels from our crowdsourced annotations. We also analyze **AUC-ROC** to measure how well the model separates the two classes across all possible decision thresholds rather than just one fixed cutoff. This is crucial because our binary labels were generated by splitting headlines at the median emotionality score, which is somewhat arbitrary. Thus, a model that ranks headlines correctly across the full score range is more meaningful than one that merely clears a single threshold.

In the time-series application we will report mean scores per outlet per year with 95% confidence intervals, supplemented by topic-stratified trends as a robustness check. If trends disappear after conditioning on topic, that is evidence of coverage-mix shift rather than genuine editorial drift. Throughout, the TF-IDF baseline serves as the performance floor — any transformer that does not substantially exceed it would not justify its computational cost for this domain.

F. Hypothesis and Expected Results

Our primary hypothesis is that a dual-head DistilBERT model trained with active learning annotations will outperform both the TF-IDF baseline and a single-task RoBERTa model on ideology classification, and that the joint training signal from emotionality annotations will improve rather than hurt ideology F1—consistent with the multi-task findings of Spinde et al. (2022). Applied to the dated Kaggle corpus, we further hypothesize that the resulting scores will reveal measurable outlet-level temporal drift in both ideology and emotionality between 2013 and 2022 that the TF-IDF baseline fails to detect.

Secondary hypotheses on the time-series outputs include the following: (1) emotionality scores increase more sharply in the 2016–2020 period coinciding with the Trump presidency; (2) ideology drift is larger on political and social topics than economic or international coverage; (3) headlines proximate to major elections show elevated scores across all outlets.

We remain open to negative results. If the multi-task model does not outperform single-task baselines, that finding is itself informative about whether ideology and emotionality share useful representational structure in short headline text. If time-series trends disappear after topic-stratification, that is a meaningful null result about the nature of apparent media drift.

IV. Preliminary Results and Analysis

A. Results So Far

We have completed two of our three planned model runs: the TF-IDF + Logistic Regression baseline and the DistilBERT ideology-only ablation. The full dual-head multi-task model will be added once we are able to collect the emotionality annotations from classmates.

- Baseline 1 — TF-IDF + Logistic Regression (Test Macro F1: 0.33)
 - The baseline performs only marginally above chance, which for a balanced three-class problem would be approximately 0.25–0.33. The model identifies left-leaning headlines reasonably well ($F1 = 0.57$) but essentially fails on center headlines ($F1 = 0.10$), consistent with our expectation that neutral framing is indistinguishable from either extreme using bag-of-words methods. The confusion matrix confirms this, as 444 center headlines are misclassified as left and 154 as right, indicating the model does not have a reliable signal for political neutrality. The large gap between training accuracy (0.78) and test accuracy (0.44) also indicates overfitting to training vocabulary—a known weakness of TF-IDF on domain-shifted text.
- Ablation — DistilBERT Ideology-Only (Test Macro F1: 0.36, Kappa: 0.12)
 - After one epoch, DistilBERT substantially outperforms TF-IDF baseline and shows meaningful improvement across all three classes. Left classification remains strongest ($F1 = 0.60$), and center and right headlines are detected at a non-trivial rate ($F1 = 0.29$ and 0.46 respectively). The center-class collapse that dominates the TF-IDF results has partially resolved, with 149 center headlines now correctly classified compared to only 40 in the baseline. Cohen's Kappa rises to 0.18, indicating the model is learning genuine ideological signal beyond majority-class prediction. Training loss (see below) decreases steadily across epochs ($1.03 \rightarrow 0.95 \rightarrow 0.86$) while validation F1 improves each epoch ($0.40 \rightarrow 0.42 \rightarrow 0.45$), confirming the model is converging rather than overfitting.



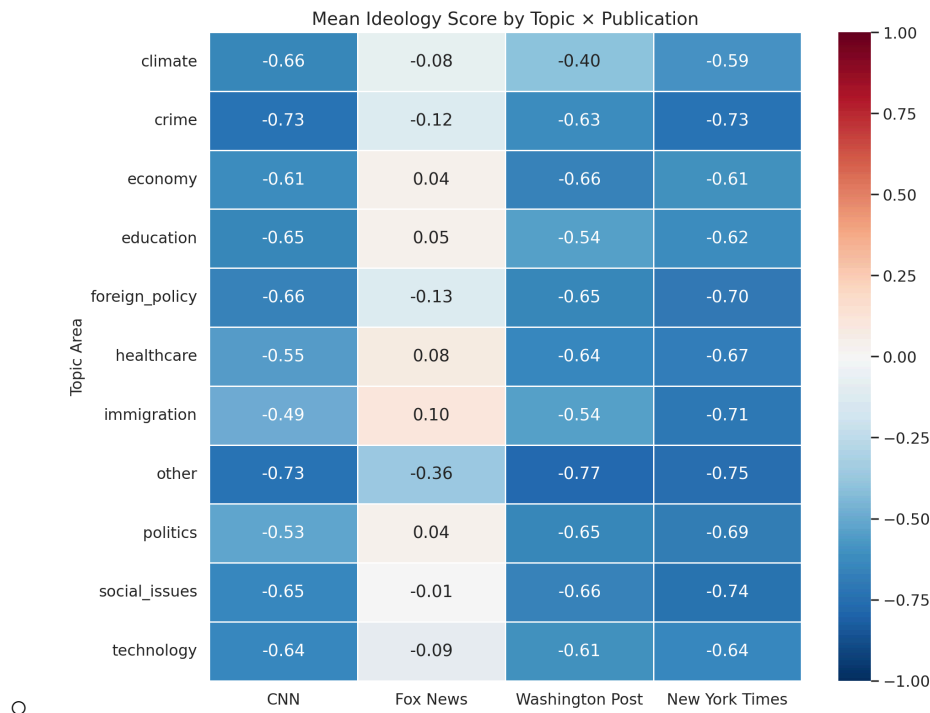
B. What These Results Mean

The preliminary results confirm the motivation for our approach. TF-IDF's failure on center headlines demonstrates that ideological framing is not recoverable from word frequencies alone. DistilBERT's gain after three epochs (12-point improvement in Macro F1) establishes that transformer-based contextual embeddings capture ideological framing that bag-of-words methods miss. Both results underscore the need for the emotionality supervision signal from our active learning annotations, which we expect will further improve performance by regularizing the shared backbone toward a richer representation of headline language.

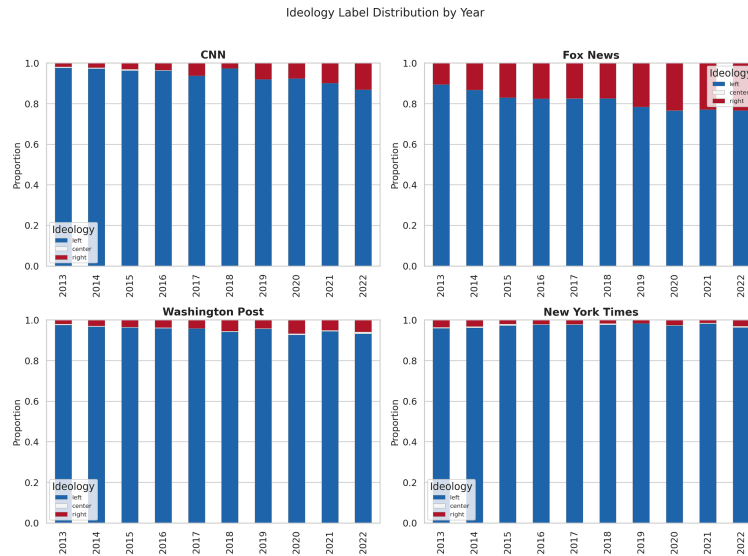
C. Commentary on (Rough Draft) Graphs

We have rough outlines for the type of plots we plan to use, although given that we have not yet done emotionality scoring, we comment in this section only on the findings of the most appropriate for where we are at in our modeling process.

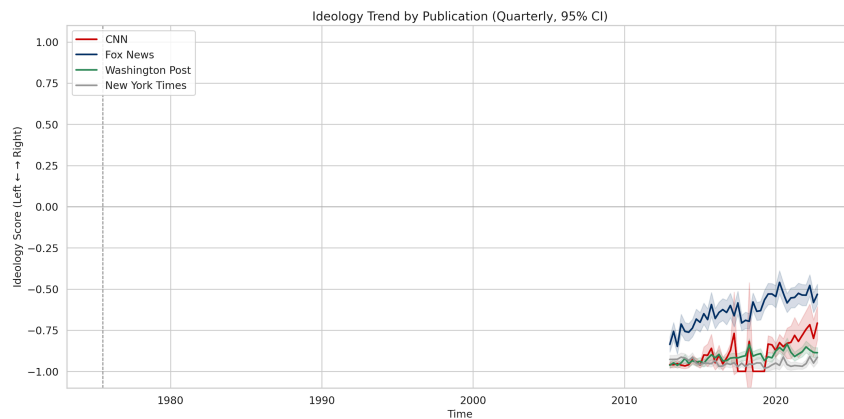
- Figure 1. Mean Ideology Score by Topic x Publication
 - Fox News crosses into positive territory on several topics — economy (+0.04), education (+0.05), healthcare (+0.08), immigration (+0.10), and politics (+0.04). This suggests the model is directionality correct in classifying Fox News headlines as right-leaning on these topics. CNN, Washington Post, and New York Times remain negative across all topics, consistent with their left-leaning labels in the training data. The spread between Fox and the other three outlets is meaningful, and the topic-level variation is plausible: immigration and healthcare show the largest Fox divergence, consistent with the real-world ideological salience of those topics.



- Figure 2. Ideology Label Distribution by Year
 - Fox News shows substantial right-leaning proportions across all years, growing noticeably from 2019 onward and reaching near-parity between left and right by 2021–2022. CNN shows increased right classification in 2018 and a notable spike in 2019 before returning to predominantly left, which is worth investigating further. Washington Post and New York Times remain predominantly left throughout, with modest right proportions that grow slightly in later years. We anticipate that these patterns will sharpen once we add emotionality supervision.



- Figure 3. Ideology Trend by Publication



- We need to fix the axis on this, but in the 2013-2022 window, the outlet divergence is striking. Fox (blue) trends strongly upward toward zero, while Washington Post and New York Times remain clustered near -0.75 throughout. CNN shows high volatility — particularly a dramatic spike toward positive values around 2019–2020. This warrants further investigation and may reflect a genuine coverage shift during that period. The upward drift in Fox and the relative stability of the other three outlets is consistent with our core hypothesis about outlet-level divergence over time.

V. Next Steps and Ablation Plan

Our immediate priorities are (1) to collect emotionality annotations from classmates and train the full dual-head model, and (2) to potentially increase training to 5 epochs to see if improvements continue over more epochs. Beyond these changes, our next steps and ablation plan includes the following:

1. Ablation 1 — Joint training effect: Compare the full dual-head model against the ideology-only ablation already trained. Any F1 improvement is directly attributable to emotionality supervision — the core test of our hypothesis.
2. Ablation 2 — Backbone vs. multi-task: Train RoBERTa single-task on ideology only. If RoBERTa matches or exceeds our dual-head DistilBERT, backbone quality dominates and multi-task learning adds little value for this domain — a meaningful finding either way.
3. Inference scaling: Because we have extremely rich data with over 1 million headlines, we currently generate ideology and emotionality scores for only 100k randomly sampled headlines due to time constraints. We plan to work with Hannah, our TA, to determine the best path forward to address these constraints.
4. Topic-stratified trends: Compute ideology and emotionality scores separately per topic area across years, controlling for the confound that apparent temporal drift may simply reflect outlets covering more polarizing topics over time rather than genuine editorial shift.
5. Election proximity: Segment the Kaggle corpus into election windows (± 90 days around 2014, 2016, 2018, 2020, 2022) and compare scores within vs. outside those windows, testing whether electoral context drives emotionality independently of outlet-level policy.

References

- [1] Lyu, H. et al. (2024). "Study of Headlines Shows Media Bias Is Growing." MEDiate Workshop @ ICWSM. University of Rochester.
- [2] Rodrigo-Ginés, F. J., Carrillo-de-Albornoz, J., & Plaza, L. (2024). "A systematic review on media bias detection." *Expert Systems with Applications*, 237, 121641.
- [3] Spinde, T., Plank, M., Krieger, J., Ruas, T., Gipp, B., & Aizawa, A. (2022). "Neural media bias detection using distant supervision with BABE." *arXiv:2209.14557*.
- [4] D'Alonzo, S. & Tegmark, M. (2022). "Machine-learning media bias." *PLOS ONE*, 17(8), e0271947.
- [5] Castillo-Campos, M. et al. (2025). "Automated detection of media bias using AI and NLP: A systematic review." *Social Science Computer Review*.
- [6] Goldsmiths, University of London (2024). "SENS-HEAD: Sensationalism detection in news headlines." Research repository, gold.ac.uk.
- [7] Scientific Reports (2025). "Clickbait detection in news headlines using RoBERTa-Large language model and deep embeddings." *Nature Portfolio*.
- [8] Bridging Human and Model Perspectives (2025). "A comparative analysis of political bias detection using LLMs." *arXiv:2511.14606*.
- [9] EMNLP (2024). "Quantifying Generative Media Bias with a Corpus of Real-world and AI-generated Articles." *Findings of ACL: EMNLP 2024*.
- [10] Spinde, T., Hamborg, F., & Gipp, B. (2023). "Media Bias Taxonomy." Systematic review of 3,140 papers, 2019–2022. media-bias-research.org.
- [11] Tracking Political Bias in U.S. News Media (2025). "Daily sentiment analysis of the 2024 presidential election using RoBERTa." *ScienceDirect / Procedia CS*.

[12] Measuring Political Bias in LLMs (2024). ACL 2024 Long Paper. Association for Computational Linguistics.

[13] Analyzing Sensationalism in News on Twitter (2024). "Clickbait journalism by legacy vs. online-native outlets." Digital Journalism.